Containers over a Finite Field

first definitions, two theorems, and one refuted conjecture

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Abstract

A container $S \triangleleft P$ (Abbott–Altenkirch–Ghani) is a set of shapes together with a set of positions for each shape; its extension $[\![S \triangleleft P]\!]X = \sum_s X^{Ps}$ is a polynomial functor, and every strictly positive type arises this way. We observe that a container is a piece of $\mathbb{F}_1$-*linear algebra*: positions form a set, and the symmetry group of n positions is $S_n = \mathrm{GL}_n(\mathbb{F}_1)$. This document deforms that structure to a general finite field.

The naive internalisation fails — **Vect** is additive, so cartesian-polynomial functors on it collapse to linear ones — and the correct move is to observe that a container is a *set-indexed coproduct of corepresentables* and to change only the base category. The resulting notion, the *q-container*, has positions that are $\mathbb{F}_q$-vector spaces and reindexing that is linear.

We prove a representation theorem (Theorem 4.1), prove that linearising the shapes destroys the theory completely (Theorem 5.2), identify the derivative with the Jackson q-derivative (Theorem 6.4), and *refute* our own conjecture that q-containers carry two derivatives (§7) — the refutation locating the interesting deformation not in q but in the self-duality of the base category. All results are computationally verified by exact enumeration over $\mathbb{F}_2$ and $\mathbb{F}_3$; §11 records exactly which claims are proved, which are verified, and which remain open.

Companion document: `finite-field-duality.pdf`, which supplies the background on forms over $\mathbb{F}_q$, the topos/abelian dichotomy, and the $q \to 1$ limit; references of the form "FFD §n" point there.

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1 Containers, and the observation that starts this

Definition 1.1 (Container)**.** A *container* is a pair $S \triangleleft P$ where S is a set of *shapes* and $P : S \to \mathbf{Set}$ assigns to each shape a set of *positions*. Its *extension* is the functor $\mathbf{Set} \to \mathbf{Set}$

$$[\![S \triangleleft P]\!]X \;=\; \sum_{s:S} (Ps \to X) \;=\; \sum_{s:S} X^{Ps},$$

"choose a shape, then label every position with an element of X".

Definition 1.2 (Container morphism)**.** A morphism $(S \triangleleft P) \to (T \triangleleft Q)$ is a pair

$$u : S \to T, \qquad f : \prod_{s:S} \big(Q(u\,s) \to Ps\big) :$$

forward on shapes, *backward* on positions. Its extension sends $(s, \varphi) \mapsto (u\,s,\ \varphi \circ f_s)$.

Theorem 1.3 (Representation theorem, AAG)**.** *The extension functor* $\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ *is full and faithful: container morphisms are exactly the natural transformations between the extensions.*

Containers are closed under $+$, $\times$, composition, Σ, Π and fixpoints, so every strictly positive type is a container; the derivative $\partial(S \triangleleft P) = \big(\sum_s Ps\big) \triangleleft \lambda(s,p).\, Ps \setminus \{p\}$ gives one-hole contexts and Huet's zippers.

Remark 1.4 (What container morphisms cannot express)*.* A morphism supplies *one* reindexing per shape, applied uniformly to every labelling of that shape. Sorting chooses its permutation by inspecting values, so it is not natural in X; by Theorem 1.3 no sorting algorithm is a container morphism. This is a proof, not an intuition, and it is worth keeping in mind as the guide to what the formalism does and does not see.

1.1 Containers are $\mathbb{F}_1$-linear algebra

Quotient containers, in which a group acts on the positions of each shape, are Joyal's analytic functors — species. The symmetry group of n positions is S_n. And $S_n = \mathrm{GL}_n(\mathbb{F}_1)$ (FFD §16). So:

> A shape with a *set* of positions is a shape with an $\mathbb{F}_1$-vector space of positions. The positions have no addition, only relabelling. Container theory is the type theory of the field with one element.

The question of this document is what happens at a general q.

2 The obstruction: one cannot internalise naïvely

Warning 2.1. There is no useful notion of "container internal to $\mathbf{Vect}_{\mathbb{F}_q}$".

By Gambino–Kock, containers are the polynomial functors of a locally cartesian closed category, built from Σ, Π and pullback. **Vect** is not locally cartesian closed; it is additive, so finite products coincide with coproducts. In **Set**, X^{Ps} is a product of $|Ps|$ copies of X; in **Vect** the product of n copies of V is $V^{\oplus n}$, so

$$\sum_s V^{\prod Ps} \;=\; \bigoplus_s V^{\oplus |Ps|}$$

is *linear* in V. Every polynomial degree above one is destroyed.

Remark 2.2. This is the topos/abelian dichotomy of FFD §11: no nontrivial category is both a topos and abelian. Containers are topos-side technology — disjoint coproducts, $\Sigma \dashv \Pi$, subobject lattices that are Boolean. **Vect** is on the abelian side. The deformation must therefore change something other than the ambient category, and identifying *what* is the whole design problem.

3 The right formulation, and the definition

Read the extension again and notice what it actually is. Writing $\mathsf{y}^W = \mathrm{Hom}(W, -)$ for the covariant functor corepresented by W,

$$[\![S \triangleleft P]\!] \;=\; \sum_{s:S} \mathsf{y}^{Ps}.$$

> **A container is a set-indexed coproduct of corepresentable functors.** Theorem 1.3 is then the Yoneda lemma: natural transformations $\mathsf{y}^W \Rightarrow \mathsf{y}^{W'}$ are maps $W' \to W$, which is exactly why container morphisms run backwards on positions.

That formulation says nothing about **Set** being the base. Change the base.

Definition 3.1 (q-container)**.** Fix a finite field $\mathbb{F}_q$. A *q-container* is a pair $S \triangleleft P$ with S a finite set of *shapes* and $P : S \to \mathbf{finVect}_{\mathbb{F}_q}$ assigning to each shape a finite-dimensional *position space*. Its *extension* is the functor $\mathbf{finVect}_{\mathbb{F}_q} \to \mathbf{Set}$

$$[\![S \triangleleft P]\!]V \;=\; \sum_{s \in S} \mathrm{Hom}_{\mathbb{F}_q}(Ps,\, V).$$

Definition 3.2 (q-container morphism)**.** A morphism $(S \triangleleft P) \to (T \triangleleft Q)$ is a pair

$$u : S \to T, \qquad f \in \prod_{s \in S} \mathrm{Hom}_{\mathbb{F}_q}\big(Q(u\,s),\, Ps\big),$$

the f_s being *linear* maps, backwards on positions. Composition is composition of the u's and (contravariantly) of the f's. Write $\mathbf{Cont}_q$ for the resulting category.

Labelling a shape now means giving a *linear map* from its position space into V: positions can be superposed, scaled and linearly recombined, and reindexing must respect that.

Remark 3.3 (Skeleton). $\mathbf{finVect}_{\mathbb{F}_q}$ is skeletal up to isomorphism on dimensions, so a q-container is determined up to isomorphism by the multiset of dimensions $(\dim Ps)_{s \in S}$. All computations below use this presentation.

Example 3.4. (a) $S = \{*\}$, $P* = \mathbb{F}_q^0$: the constant functor 1.
(b) $S = \{*\}$, $P* = \mathbb{F}_q^1$: the identity, since $\mathrm{Hom}(\mathbb{F}_q, V) \cong V$.
(c) $S = \{*\}$, $P* = \mathbb{F}_q^d$: the functor $V \mapsto V^{\oplus d}$ — *not* $V^{\otimes d}$ (see Warning 2.1).
(d) $S = \mathbb{N}$, $Pn = \mathbb{F}_q^n$: the "q-list" container, $[\![-]\!]V = \coprod_n \mathrm{Hom}(\mathbb{F}_q^n, V)$.

Proposition 3.5 ($q \to 1$)**.** *Taking* $\mathbf{Vect}_{\mathbb{F}_1} = \mathbf{FinSet}_*$ *with* $\mathrm{Hom}_{\mathbb{F}_1}$ *the pointed maps (FFD §16), Definitions 3.1 and 3.2 specialise to pointed containers and pointed container morphisms.*

4 Theorem A: the representation theorem survives

Theorem 4.1. *The extension functor* $\mathbf{Cont}_q \to [\mathbf{finVect}_{\mathbb{F}_q}, \mathbf{Set}]$ *is full and faithful. Explicitly, with* $d_s = \dim Ps$ *and* $e_t = \dim Qt$*,*

$$\mathbf{Cont}_q(C, D) \;\cong\; \mathrm{Nat}\big([\![C]\!], [\![D]\!]\big) \;\cong\; \prod_{s \in S} \sum_{t \in T} \mathrm{Hom}_{\mathbb{F}_q}(Qt, Ps),$$

$$\textit{so that} \qquad \big|\mathbf{Cont}_q(C, D)\big| \;=\; \prod_s \sum_t q^{\,e_t d_s}.$$

Proof. Colimits in a functor category are pointwise, so Σ is the coproduct in $[\mathbf{finVect}_q, \mathbf{Set}]$ and $\mathrm{Nat}\big(\sum_s F_s, G\big) \cong \prod_s \mathrm{Nat}(F_s, G)$. By Yoneda, $\mathrm{Nat}(\mathrm{y}^W, G) \cong G(W)$. Hence

$$\mathrm{Nat}([\![C]\!], [\![D]\!]) \;\cong\; \prod_s [\![D]\!](Ps) \;=\; \prod_s \sum_t \mathrm{Hom}(Qt, Ps).$$

On the other side, Definition 3.2 gives $\mathbf{Cont}_q(C, D) = \sum_{u : S \to T} \prod_s \mathrm{Hom}(Q(u\,s), Ps)$, and the distributive law of Π over Σ in $\mathbf{Set}$,

$$\prod_{s \in S} \sum_{t \in T} A(s, t) \;\cong\; \sum_{u : S \to T} \prod_{s \in S} A\big(s, u\,s\big),$$

identifies the two. Chasing the bijection shows it is exactly the extension map. □

Remark 4.2 (Where the proof spends its budget). The proof uses precisely two facts about the *indexing*: that Σ is a coproduct with disjoint, connected summands, and that Π distributes over Σ. Both are topos-side properties of $\mathbf{Set}$, and neither holds for $\oplus$ in $\mathbf{Vect}$. Theorem 5.2 is the consequence.

4.1 Computational verification

Verified by exact enumeration, two methods, neither assuming Yoneda: *exhaustive* search over all function families $\{\alpha_n : [\![C]\!](\mathbb{F}_q^n) \to [\![D]\!](\mathbb{F}_q^n)\}$ keeping the natural ones; and *generic*, enumerating candidates by their value on (s, id_{Ps}), verifying naturality of each in full, and separately checking that every source element is $[\![C]\!](\varphi)$ of a generic (so no natural family can escape the count).

q	C	D	max dim	method	#natural	$\lvert\mathbf{Cont}_q(C,D)\rvert$
2	(1)	(1)	1	exhaustive	2	2
2	(1)	$(1,1)$	1	exhaustive	4	4
2	$(1,1)$	(1)	1	exhaustive	4	4
2	$(0,1)$	(1)	1	exhaustive	2	2
2	$(1,1)$	$(1,1)$	1	exhaustive	16	16
3	(1)	(1)	1	exhaustive	3	3
3	(1)	$(1,1)$	1	exhaustive	6	6
2	(1)	(2)	2	exhaustive	4	4
2	(2)	(1)	2	generic	4	4
2	(2)	$(1,1)$	2	generic	8	8
2	$(1,2)$	$(2,1)$	2	generic	120	120
3	(2)	(1)	2	generic	9	9

A separate elementwise check ($q = 2$, $C = D = (1,1)$) confirms that the map from container morphisms to natural families is injective and surjective, not merely equinumerous.

5 Theorem B: linearising the shapes destroys the theory

Definition 3.1 keeps the extension **Set**-valued. Since $\mathrm{Hom}_{\mathbb{F}_q}(Ps, V)$ is itself a vector space, one is tempted to replace Σ by $\oplus$ and land in **Vect**. This is a mistake, and the failure is total rather than marginal.

Definition 5.1. The *linear extension* of $S \triangleleft P$ is the $\mathbb{F}_q$-linear functor $[\![S \triangleleft P]\!]_{\mathrm{lin}} V = \bigoplus_{s \in S} \mathrm{Hom}_{\mathbb{F}_q}(Ps, V)$.

Theorem 5.2. *For finite S and T,*

$$\mathrm{Nat}\big([\![C]\!]_{\mathrm{lin}}, [\![D]\!]_{\mathrm{lin}}\big) \;\cong\; \mathrm{Hom}_{\mathbb{F}_q}\Big(\bigoplus_{t \in T} Qt,\; \bigoplus_{s \in S} Ps\Big), \qquad \textit{of cardinality } q^{(\sum_s d_s)(\sum_t e_t)}.$$

In particular the right-hand side depends only on the total position spaces: the shape sets S and T are forgotten entirely.

Proof. A natural transformation between additive functors is automatically $\mathbb{F}_q$-linear in each component (use the scalar endomorphisms $\lambda \cdot \mathrm{id}$ for homogeneity, and the diagonal and codiagonal $V \to V \oplus V \to V$ for additivity). Enriched Yoneda gives $\mathrm{Nat}(\mathrm{Hom}(W, -), G) \cong G(W)$ for linear G, and $\oplus$ is a biproduct, hence also a product, so

$$\mathrm{Nat}\Big(\bigoplus_s \mathrm{Hom}(Ps, -), G\Big) \;\cong\; \prod_s G(Ps) \;=\; \prod_s \bigoplus_t \mathrm{Hom}(Qt, Ps)$$

$$\cong\; \prod_s \mathrm{Hom}\Big(\bigoplus_t Qt,\; Ps\Big) \;\cong\; \mathrm{Hom}\Big(\bigoplus_t Qt,\; \bigoplus_s Ps\Big). \qquad \square$$

Corollary 5.3. *The linear extension functor is* ***neither full nor faithful****.*

Example 5.4 (Both failures at once, and cardinality is no guide)**.** Take $q = 2$, $C = (1)$, $D = (1, 1)$. Then $|\mathbf{Cont}_q(C, D)| = 4$ and $|\mathrm{Nat}_{\mathrm{lin}}| = 4$ — the cardinalities agree — yet the map between them is neither injective nor surjective, having image of size 3:

- *not faithful*: the two morphisms $(u = 0, f = 0)$ and $(u = 1, f = 0)$ have the same linear extension, namely 0. In **Set** these land in different summands and stay apart; in **Vect** the zero vector is shared.
- *not full*: the "diagonal" map $(1, 1) \in \mathrm{Hom}(\mathbb{F}_2^2, \mathbb{F}_2)$ is natural but is not the extension of any container morphism, since a morphism must choose a single target shape per source shape.

q	$C \to D$	$\lvert\mathbf{Cont}_q\rvert$	image	$\lvert\mathrm{Nat}_{\mathrm{lin}}\rvert$	faithful	full
2	$(1) \to (1)$	2	2	2	yes	yes
2	$(1) \to (1, 1)$	4	3	4	**refuted**	**refuted**
2	$(1, 1) \to (1, 1)$	16	9	16	**refuted**	**refuted**
2	$(1) \to (1, 1, 1)$	6	4	8	**refuted**	**refuted**
3	$(1) \to (1, 1)$	6	5	9	**refuted**	**refuted**

The representation theorem survives exactly as long as the shapes stay a set. Positions may be linearised freely — that is Theorem 4.1. Linearising the shapes replaces a disjoint union by a biproduct, natural transformations may then take linear combinations across different target shapes, and by Theorem 5.2 the shape set is annihilated. This is Remark 4.2 made quantitative, and the topos/abelian dichotomy of FFD §11 in the form of a cardinality identity.

6 Counting, and the Jackson q-derivative

Proposition 6.1 (Counting series)**.** $\big|[\![S \triangleleft P]\!](\mathbb{F}_q^n)\big| = \sum_s \big(q^n\big)^{d_s}$. *Writing $x = q^n$, the counting series of a q-container is $f_C(x) = \sum_s x^{d_s}$, exactly parallel to $|[\![S \triangleleft P]\!](X)| = \sum_s |X|^{|Ps|}$ in* **Set***, under $|X| \leftrightarrow q^n$ and $|Ps| \leftrightarrow \dim Ps$.*

Verified by direct enumeration for $q = 2$, $n \le 3$, $C \in \{(1), (2), (1, 2)\}$.

Definition 6.2 (Derivative)**.** For a q-container $S \triangleleft P$, put

$$\partial(S \triangleleft P) \;=\; \Big(\textstyle\sum_s \{\text{hyperplanes of } Ps\}\Big) \;\triangleleft\; \big((s, H) \mapsto H\big),$$

so a shape of ∂C is a shape of C together with a hyperplane of its position space, and the labelling forgets the missing direction.

Definition 6.3 (Jackson q-derivative)**.** $D_q f(x) = \dfrac{f(qx) - f(x)}{(q-1)x}$, so that $D_q x^d = [d]_q\, x^{d-1}$ with $[d]_q = 1 + q + \cdots + q^{d-1}$.

Theorem 6.4. *$f_{\partial C} = D_q f_C$. As $q \to 1$, $[d]_q \to d$, so $D_q \to d/dx$ and ∂ degenerates to the ordinary container derivative.*

Proof. $\mathbb{F}_q^d$ has $\begin{bmatrix} d \\ d-1 \end{bmatrix}_q = [d]_q$ hyperplanes, each of dimension $d - 1$. □

Verified exactly for $q \in \{2, 3\}$ and $C \in \{(1), (2), (3), (1, 2), (2, 2, 3)\}$; e.g. $q = 3$, $C = (2, 2, 3)$ gives $f_C = x^3 + 2x^2$ and $f_{\partial C} = 13x^2 + 8x = D_q f_C$.

7 A refuted conjecture, and what it was really about

Refuted conjecture 7.1. We conjectured that a q-container has *two* derivatives — ∂_{sub}, restricting to a hyperplane of Ps, and ∂_{quot}, quotienting by a line — exchanged by the polarity $W \mapsto W^{\perp}$ of FFD §14, and merging at $q = 1$. The evidence offered was that the Gaussian Pascal recursion has two forms,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \qquad \begin{bmatrix} n \\ k \end{bmatrix}_q = q^{\,n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k \end{bmatrix}_q,$$

which coincide at $q = 1$.

Proposition 7.2 (Refutation)**.** *Over* $\mathbb{F}_q$, $\partial_{\mathrm{sub}} \cong \partial_{\mathrm{quot}}$.

Proof. $\begin{bmatrix} d \\ d-1 \end{bmatrix}_q = \begin{bmatrix} d \\ 1 \end{bmatrix}_q = [d]_q$, so a shape has as many hyperplanes as lines; and $H \cong Ps/\ell \cong \mathbb{F}_q^{d-1}$ for every hyperplane H and line ℓ. Since a q-container remembers only the dimension of each position space, the two constructions give isomorphic q-containers. □

Confirmed computationally for $q \in \{2, 3\}$ and all position dimensions ≤ 3. So the two Pascal recursions are not two derivatives; they are *one derivative counted two ways.*

Remark 7.3 (The diagnosis). The conjecture silently assumed a base category in which subobjects and quotients are not interchangeable. $\mathbf{Vect}_{\mathbb{F}_q}$ is semisimple: every short exact sequence splits, and the category is self-dual. The interesting deformation is therefore not q but the *failure of self-duality of the base.* Note that this is a different failure of splitting from the one in FFD Part I: there, a *form* on a semisimple category fails to split $V = W \perp W^{\perp}$; here it is the *category itself* that splits too well. Conflating them was the error.

Conjecture 7.4 (Repaired)**.** *Let* $\mathcal{A}$ *be a finite* $\mathbb{F}_q$*-linear abelian category and define* $\mathcal{A}$*-containers by Definition 3.1 with* $P : S \to \mathcal{A}$*. Then* $\partial_{\mathrm{sub}} \cong \partial_{\mathrm{quot}}$ *if and only if* $\mathcal{A}$ *is self-dual. In particular they separate over* $\mathcal{A} = \mathrm{Rep}_{\mathbb{F}_q}(\vec{A}_n)$*.*

Example 7.5 (Smallest witness, hand-checked)**.** Let $\mathcal{A} = \mathrm{Rep}_{\mathbb{F}_q}(1 \to 2)$, with simples $S_1 = (\mathbb{F}_q, 0)$, $S_2 = (0, \mathbb{F}_q)$ and projective $P_1 = (\mathbb{F}_q \xrightarrow{\mathrm{id}} \mathbb{F}_q)$. The proper nonzero subobjects of P_1 are just S_2, since a subrepresentation (W_1, W_2) needs $\mathrm{id}(W_1) \subseteq W_2$; and $P_1/S_2 = S_1$. As $S_1 \ncong S_2$,

$$\partial_{\mathrm{sub}}(P_1) = S_2 \;\neq\; S_1 = \partial_{\mathrm{quot}}(P_1).$$

Remark 7.6. One might expect the repair to the derivative and the enrichment of the composition (§8) to have a single cause, namely non-semisimplicity. They do not. Non-semisimplicity enriches the Hall algebra, but a category can be non-semisimple and still self-dual — finite modules over a discrete valuation ring are, by Matlis duality — in which case the two derivatives still coincide. The two phenomena sit on independent axes, tabulated in Table 1.

8 Composition, and where Hall–Littlewood enters

The classical theorems quoted here are Hall's and Macdonald's; their application to q-containers is design, and nothing in this section has been verified computationally. An earlier draft of this section asserted that Day convolution has Hall-algebra structure constants. That is false, and §8.2 is the correction.

8.1 Two candidate composition laws

Definition 8.1 (Day convolution)**.** Let $(\mathcal{A}, \otimes)$ be a small monoidal category. The functor category $[\mathcal{A}, \mathbf{Set}]$ carries the monoidal structure

$$(F \star G)(c) \;=\; \int^{a,b} \mathcal{A}(a \otimes b,\, c) \times F(a) \times G(b),$$

the unique one making the Yoneda embedding strong monoidal.

Example 8.2 (The species product)**.** For $\mathcal{A} = (\mathbf{FinSet}_{\mathrm{bij}}, \sqcup)$ the coend collapses to $(F \star G)[n] = \sum_{A \sqcup B = [n]} F[A] \times G[B]$, the species product, which decategorifies to multiplication of exponential generating functions.

Definition 8.3 (Hall algebra)**.** Let $\mathcal{A}$ be an abelian category in which every object has finitely many subobjects and all Hom and Ext^1 sets are finite. Its *Hall algebra* $H(\mathcal{A})$ is the free $\mathbb{Z}$-module on isomorphism classes, with

$$[M] \cdot [N] \;=\; \sum_{[L]} g^{L}_{MN}\, [L], \qquad g^{L}_{MN} \;=\; \#\big\{\, U \subseteq L \;:\; U \cong N,\; L/U \cong M \,\big\}.$$

So g^{L}_{MN} counts *subobjects with prescribed quotient* — that is, all extensions of M by N with middle term L, split or not.

These are genuinely different operations, and it is worth computing both in the smallest case before relying on either.

Proposition 8.4 (Day convolution over $\mathbb{F}_q$)**.** *Let $\mathcal{G}_q = \bigsqcup_n B\,\mathrm{GL}_n(\mathbb{F}_q)$ be the groupoid of finite-dimensional $\mathbb{F}_q$-spaces and isomorphisms, monoidal under $\oplus$, and let δ_k be the point-functor supported on dimension k. Then*

$$(\delta_k \star \delta_{n-k})(\mathbb{F}_q^n) \;=\; \mathrm{GL}_n \,/\big(\mathrm{GL}_k \times \mathrm{GL}_{n-k}\big), \qquad \textit{of cardinality } \begin{bmatrix} n \\ k \end{bmatrix}_q q^{\,k(n-k)} :$$

the number of ordered direct-sum decompositions of $\mathbb{F}_q^n$ into subspaces of dimensions k and $n-k$.

Example 8.5. $n = 2$, $k = 1$, $q = 2$: the formula gives $3 \cdot 2 = 6$, and directly, $\mathbb{F}_2^2$ has 3 lines, each with 2 complements.

Proposition 8.6 (The Hall product over $\mathbb{F}_q$)**.** *For $\mathcal{A} = \mathbf{Vect}_{\mathbb{F}_q}$, $g^{n}_{k,\,n-k} = \begin{bmatrix} n \\ k \end{bmatrix}_q$, the number of subspaces of the given dimension. In the same case as above this is* 3*, not* 6*.*

Corollary 8.7 (Agreement, up to a twist, when the base is semisimple)**.** *Over $\mathbf{Vect}_{\mathbb{F}_q}$ the two structure constants differ by the factor $q^{\,k(n-k)} = q^{\langle M,N\rangle}$, where $\langle M, N\rangle = \dim \mathrm{Hom}(M,N) - \dim \mathrm{Ext}^1(M,N)$ is the Euler form — the number of complements to a fixed subspace. This is exactly the twist relating the Hall algebra to its twisted form, so over a semisimple base the discrepancy is cosmetic.*

8.2 Over a filtered base they diverge essentially

Definition 8.8 (Discrete valuation ring, type of a module)**.** A *discrete valuation ring* $\mathcal{O}$ is a local principal ideal domain that is not a field; write $\mathfrak{m} = (\pi)$ and assume the residue field $\mathcal{O}/\mathfrak{m} \cong \mathbb{F}_q$ is finite. Canonical examples: $\mathbb{F}_q[[t]]$, and $\mathbb{Z}_p$ with $q = p$. Every finite $\mathcal{O}$-module is uniquely $M_\lambda \cong \bigoplus_i \mathcal{O}/(\pi^{\lambda_i})$ for a partition λ, its *type*; write $n(\lambda) = \sum_i (i-1)\lambda_i$.

Such a module is an $\mathbb{F}_q$-space with a nilpotent operator: positions carrying a filtration, a "shift with decay". Setting $\pi = 0$ recovers $\mathbf{Vect}_{\mathbb{F}_q}$, so this enriches Definition 3.1 rather than departing from it.

Example 8.9 (The computation that separates them)**.** Multiply the class of the simple module by itself.

$$\textbf{Hall:} \qquad u_{(1)} \cdot u_{(1)} \;=\; u_{(2)} \;+\; (q+1)\, u_{(1,1)}.$$

The term $u_{(2)}$ arises from the *non-split* extension $\mathcal{O}/\pi \subset \mathcal{O}/\pi^2$; the coefficient $q+1$ counts the lines in $\mathbb{F}_q^2$.

$$\textbf{Day convolution along } \oplus: \qquad \delta_{(1)} \star \delta_{(1)} \;=\; q(q+1)\, \delta_{(1,1)}, \quad \text{and nothing else.}$$

Proposition 8.10. *$(F \star G)(L)$ is empty unless $L \cong U \oplus W$ for some U, W in the supports of F and G. Hence Day convolution along $\oplus$ cannot see non-split extensions at all.*

Correction. Day convolution along $\oplus$ is the Cauchy product of q-species; its structure constants count *splittings*, and agree with the twisted Hall constants only when the base is semisimple. It is *not* the Hall product. Reaching the Hall algebra requires convolution along the span of short exact sequences,

$$\mathcal{A} \times \mathcal{A} \;\longleftarrow\; \mathcal{E} \;\longrightarrow\; \mathcal{A}, \qquad \big(0 \to N \to L \to M \to 0\big) \;\longmapsto\; \big((M,N),\, L\big),$$

pushing forward along the middle-term map — the Ringel–Hall product as a convolution algebra on the stack of objects (Kapranov–Vasserot; Schiffmann's lectures). Day convolution deletes precisely the term that carries the content.

§9 identifies what the two products actually are: convolution along $\oplus$ induces from the *Levi*, the Hall product induces from the *parabolic*, and the discrepancy is the unipotent radical.

8.3 The Hall side is Hall–Littlewood

Definition 8.11 (Hall–Littlewood polynomials)**.** $P_\lambda(x;t)$ is the basis of the ring Λ of symmetric functions that is monic–triangular against the monomial basis for dominance order and orthogonal for the t-deformed Hall inner product. It interpolates: $P_\lambda(x;0) = s_\lambda$ and $P_\lambda(x;1) = m_\lambda$.

Theorem 8.12 (Hall; Macdonald, *Symmetric Functions and Hall Polynomials*, II.4 and III.3)**.** *Let $\mathcal{A}_\mathcal{O}$ be the category of finite modules over a discrete valuation ring with residue field $\mathbb{F}_q$. The structure constants $g^\lambda_{\mu\nu}(q)$ of Definition 8.3 are polynomials in q with integer coefficients — the Hall polynomials — independent of which $\mathcal{O}$ realises the residue field, and*

$$H(\mathcal{A}_\mathcal{O}) \;\xrightarrow{\;\sim\;}\; \Lambda, \qquad u_\lambda \;\longmapsto\; q^{-n(\lambda)}\, P_\lambda\big(x;\, q^{-1}\big)$$

is a ring isomorphism. Equivalently $f^\lambda_{\mu\nu}(t) = t^{\,n(\lambda)-n(\mu)-n(\nu)} g^\lambda_{\mu\nu}(t^{-1})$ are the Hall–Littlewood structure constants.

Example 8.13 (Example 8.9 pushed through)**.** With $n((1)) = n((2)) = 0$ and $n((1,1)) = 1$, the Hall identity becomes

$$P_1^2 \;=\; P_{(2)} \;+\; \big(1 + q^{-1}\big) P_{(1,1)} \;=\; P_2 + (1+t)\, P_{11}.$$

At $t = 0$ this is $s_1^2 = s_2 + s_{11}$; at $t = 1$ it is $m_1^2 = m_2 + 2m_{11}$. Both correct. Note that the content sits in the P_2 term — the non-split extension — which is exactly what Proposition 8.10 deletes.

Weakened conclusion. q-containers with $\mathcal{O}$-module positions admit *two* candidate composition laws. Day convolution along $\oplus$ gives the Cauchy product; convolution along short exact sequences gives the Hall product and hence, by Theorem 8.12, multiplication of Hall–Littlewood polynomials $P_\lambda(x;q^{-1})$. The two agree up to the Euler twist over $\mathbf{Vect}_{\mathbb{F}_q}$ and disagree essentially the moment the positions carry a filtration. Which one deserves the name "composition of containers" is open — see Question 8.16.

Remark 8.14 (Green's polynomials close the circle). The same $P_\lambda(x;q^{-1})$ carry the character theory of the groups whose $\mathbb{F}_1$-shadow started this document: $p_\mu = \sum_\lambda Q^\lambda_\mu(q)\, P_\lambda(x;q^{-1})$ with $Q^\lambda_\mu(q) \in \mathbb{Z}[q]$ the unipotent character values of $\mathrm{GL}_n(\mathbb{F}_q)$, and $Q^\lambda_\mu(1) = \chi^\lambda_\mu$, the character table of S_n. Composing with Kostka–Foulkes, $s_\lambda = \sum_\nu K_{\lambda\nu}(q) P_\nu(x;q)$, factors this as $Q = \chi^{\mathsf{T}} K$. So the Hall–Littlewood basis is the coordinate system in which Zelevinsky's tower $\bigoplus_n \mathrm{Rep}(\mathrm{GL}_n(\mathbb{F}_q))$ — degenerating to $\bigoplus_n \mathrm{Rep}(S_n)$, hence to species — is legible.

8.4 Two independent axes

It is tempting to say that "non-semisimple base" is the single repair that both enriches the composition and separates the two derivatives of §7. That is false, and the distinction matters.

Definition 8.15 (Self-dual category)**.** $\mathcal{A}$ is *self-dual* if there is an exact anti-equivalence $\mathcal{A}^{\mathrm{op}} \simeq \mathcal{A}$, that is, a duality exchanging subobjects and quotients. $\mathbf{Vect}_{\mathbb{F}_q}$ is self-dual by $V \mapsto V^*$; $\mathcal{A}_\mathcal{O}$ is self-dual by Matlis duality $M \mapsto \mathrm{Hom}_\mathcal{O}(M, K/\mathcal{O})$, which preserves the type λ; $\mathrm{Rep}_{\mathbb{F}_q}(\vec{A}_2)$ is *not*, as $S_1 \not\cong S_2$ witnesses.

base $\mathcal{A}$ for positions	Hall algebra	self-dual?	∂_{sub} vs ∂_{quot}
$\mathbf{Vect}_{\mathbb{F}_q}$	divided powers, $\begin{bmatrix} n \\ k \end{bmatrix}_q$	yes	coincide (§7, proved)
finite $\mathcal{O}$-modules	Λ, Hall–Littlewood $P_\lambda(x;q^{-1})$	yes (Matlis)	coincide (predicted)
$\mathrm{Rep}_{\mathbb{F}_q}(\vec{A}_2)$	$U_q(\mathfrak{n}^-)$ for $\mathfrak{sl}_2$, noncomm.	**no**	**separate** (Conj. 7.4)

Table 1: Filtration depth and failure of self-duality are independent. The middle row is non-semisimple, has a rich commutative Hall algebra, and still has coincident derivatives; only the bottom row separates them.

So: *filtration depth* buys Hall–Littlewood; *failure of self-duality* buys two derivatives. Neither implies the other.

8.5 What is now open, and what to compute

Question 8.16. q-containers are the coproducts of corepresentables. Is that class closed under Hall convolution? Day convolution along $\oplus$ preserves it, because $\mathsf{y}^U \star \mathsf{y}^W \cong \mathsf{y}^{U\oplus W}$. The Hall product has no such easy answer, and if the class is not closed then "Hall–Littlewood composition" is a statement about q-species, not about q-containers.

1. Settle Question 8.16. This is the load-bearing one: it decides whether §8 is about containers at all.
2. Enumerate both products for $\mathcal{O} = \mathbb{F}_2[t]/(t^k)$, $k \le 3$, and confirm Example 8.9 and the Hall polynomials $g^\lambda_{\mu\nu}(q)$ for $|\lambda| \le 4$.
3. Confirm the middle row of Table 1: that Matlis duality alone forces $\partial_{\mathrm{sub}} \cong \partial_{\mathrm{quot}}$, isolating failure of self-duality as the sole obstruction.
4. Check that Theorem 4.1 is stable under change of base; the proof used only that the shapes form a set.

9 Induction, restriction, and change of base

The two most useful operations in representation theory are induction and restriction, and they are adjoint. This section shows that any such adjunction is already native to the container calculus, and that doing so names the discrepancy of §8.2.

9.1 Adjunctions lift, with a variance flip

Definition 9.1 (Change of base)**.** For a functor $A : \mathcal{A} \to \mathcal{B}$ and a container $C = S \triangleleft P$ with positions in $\mathcal{A}$, write $A_*C = S \triangleleft (A \circ P)$, a container with positions in $\mathcal{B}$. On morphisms, $A_*(u, f) = (u, Af)$.

Theorem 9.2. *Let $A : \mathcal{A} \to \mathcal{B}$ and $B : \mathcal{B} \to \mathcal{A}$ with $B \dashv A$. Then $A_* \dashv B_*$ on containers:*

$$\mathbf{Cont}_{\mathcal{B}}\big(A_*C,\, D\big) \;\cong\; \mathbf{Cont}_{\mathcal{A}}\big(C,\, B_*D\big).$$

Proof. By Theorem 4.1 both sides are products of sums of hom-sets:

$$\mathbf{Cont}_{\mathcal{B}}(A_*C, D) = \prod_s \sum_t \mathrm{Hom}_{\mathcal{B}}\big(Qt,\, A(Ps)\big),$$
$$\mathbf{Cont}_{\mathcal{A}}(C, B_*D) = \prod_s \sum_t \mathrm{Hom}_{\mathcal{A}}\big(B(Qt),\, Ps\big),$$

and $\mathrm{Hom}_{\mathcal{B}}(X, AY) \cong \mathrm{Hom}_{\mathcal{A}}(BX, Y)$ is exactly $B \dashv A$. The bijection is natural in each factor, hence in C and D. □

Remark 9.3 (Why the variance flips)*.* The *left* adjoint on positions becomes the *right* adjoint on containers, and conversely. This is forced by Definition 3.2: container morphisms run backwards on positions, so an adjunction used on the position category is consumed in the opposite slot.

Proposition 9.4 (Extensions)**.** *If $A \dashv B$ then $[\![A_*C]\!] = [\![C]\!] \circ B$. Pushing positions along a left adjoint is precomposing the extension with its right adjoint.*

Proof. $[\![A_*C]\!](W) = \sum_s \mathrm{Hom}_{\mathcal{B}}(A(Ps), W) \cong \sum_s \mathrm{Hom}_{\mathcal{A}}(Ps, BW) = [\![C]\!](BW)$. □

9.2 Induction and restriction

Definition 9.5. For $H \le G$ and a field k: $\mathrm{Res}^G_H : \mathrm{Rep}_k(G) \to \mathrm{Rep}_k(H)$ restricts the action; $\mathrm{Ind}^G_H = k[G] \otimes_{k[H]} -$ and $\mathrm{Coind}^G_H = \mathrm{Hom}_{k[H]}(k[G], -)$. One has $\mathrm{Ind} \dashv \mathrm{Res} \dashv \mathrm{Coind}$, and for $[G : H] < \infty$, $\mathrm{Ind} \cong \mathrm{Coind}$, so Res is *biadjoint* to Ind.

Corollary 9.6. *For $H \le G$ of finite index, positions in $\mathrm{Rep}_k(-)$ give*

$$\mathrm{Ind}_* \;\dashv\; \mathrm{Res}_* \;\dashv\; \mathrm{Ind}_*,$$

and, by Proposition 9.4,

$$[\![\mathrm{Ind}_*\, C]\!] = [\![C]\!] \circ \mathrm{Res}, \qquad [\![\mathrm{Res}_*\, C]\!] = [\![C]\!] \circ \mathrm{Ind}\,.$$

Inducting the positions is restricting the values, and conversely: Frobenius reciprocity holds shape by shape. Counting, with $k = \mathbb{F}_q$,

$$\big|\mathbf{Cont}_G(C, D)\big| \;=\; \prod_s \sum_t q^{\,\dim \mathrm{Hom}_G(Qt,\, Ps)},$$

and in the semisimple case $\dim \mathrm{Hom}_G(M, N) = \langle \chi_M, \chi_N \rangle$: the container hom-set is computed by inner products of characters. Taking G trivial recovers Theorem 4.1.

9.3 Parabolic versus Levi: what §8.2 was about

Let $G = \mathrm{GL}_n(\mathbb{F}_q)$, let P be the parabolic stabilising a k-dimensional subspace, $L = \mathrm{GL}_k \times \mathrm{GL}_{n-k}$ its Levi factor and U its unipotent radical, so $P = L \ltimes U$.

Definition 9.7 (Harish-Chandra induction)**.** $R_L^G = \mathrm{Ind}_P^G \circ \mathrm{Infl}_L^P : \mathrm{Rep}(L) \to \mathrm{Rep}(G)$: inflate along $P \twoheadrightarrow L$, killing U, then induce. Its adjoint ${}^*R_L^G$ takes U-invariants and restricts to L.

Proposition 9.8.

$$[G : P] = \begin{bmatrix} n \\ k \end{bmatrix}_q, \qquad [G : L] = \begin{bmatrix} n \\ k \end{bmatrix}_q q^{k(n-k)}, \qquad [P : L] = |U| = q^{k(n-k)}.$$

operation	induces from	index	container side
Harish-Chandra $R_L^G = \mathrm{Ind}_P^G \mathrm{Infl}$	parabolic P	$\begin{bmatrix} n \\ k \end{bmatrix}_q$	Hall product
ordinary Ind_L^G	Levi L	$\begin{bmatrix} n \\ k \end{bmatrix}_q q^{k(n-k)}$	Day conv. along $\oplus$

The discrepancy of §8.2 is the unipotent radical. Day convolution along $\oplus$ induces from the *Levi*: it remembers a choice of complement, which is not canonical, and the redundancy is exactly $|U| = q^{k(n-k)}$. Harish-Chandra induction goes through the *parabolic* and inflates U away, which is what lets it see extensions rather than splittings. The factor I could previously only describe as an Euler-form twist is the order of the unipotent radical.

Example 9.9 ($\mathrm{GL}_2(\mathbb{F}_2)$)**.** $|G| = 6$; L is trivial so $[G : L] = 6$, the ordered pairs of complementary lines; $P = B$ has order 2, so $[G : P] = 3$, the lines; and $|U| = 2 = q^1$.

Remark 9.10 (Zelevinsky, and $q = 1$). Taking induction and restriction to be *Harish-Chandra* induction and restriction along Levi factors and their unipotent subgroups is exactly what gives $\bigoplus_n \mathrm{Rep}(\mathrm{GL}_n(\mathbb{F}_q))$ its structure as a positive self-adjoint Hopf algebra, decomposing as a tensor product of copies of the elementary one indexed by cuspidal representations (Zelevinsky). At $q = 1$ there is no unipotent radical to inflate away, the two rows of the table coincide, and one recovers the induction product $\mathrm{Ind}_{S_k \times S_{n-k}}^{S_n}$ on $\bigoplus_n \mathrm{Rep}(S_n)$ — which is precisely the species product, i.e. Day convolution along $\sqcup$. The parabolic/Levi distinction is a q-phenomenon, and it vanishes for the same reason the two derivatives of §7 merge.

Remark 9.11 (The Hecke algebra). $\mathrm{Ind}_B^G \mathbf{1} = k[G/B] = k[\text{complete flags}]$ is a free module on a hom-set — a container in the $k[-]$ sense of §8 — and $\mathrm{End}_G(\mathrm{Ind}_B^G \mathbf{1}) = H_n(q)$, the Hecke algebra, degenerating to $k[S_n]$ at $q = 1$. So the Hecke algebra is the endomorphism algebra of the flag container.

Question 9.12 (Mackey). The Mackey formula $\mathrm{Res} \circ \mathrm{Ind} = \bigoplus_{HgK} \mathrm{Ind} \circ \mathrm{conj} \circ \mathrm{Res}$ should lift to a distributive law between Ind_* and Res_*. It is the compatibility that makes the PSH structure work, so if it lifts cleanly the container calculus carries both operations *and* their interaction.

10 A general programme: lifting Macdonald theory to containers

10.1 Decategorification

Proposition 6.1 sends a q-container to a polynomial. That map is the bottom of a ladder:

$$\underbrace{q\text{-containers}}_{\text{functors, natural bijections}} \longrightarrow \underbrace{\text{symmetric functions}}_{\text{characters, if the positions carry a group action}}$$

$$\longrightarrow \underbrace{q\text{-series}}_{f_C(x)=\sum_s x^{\dim Ps}} \longrightarrow \text{numbers}$$

and it is a homomorphism for every operation we have:

container level	polynomial level
coproduct $C + D$	$f_C + f_D$
product $C \times D$ (Def. 10.1)	$f_C \cdot f_D$
Day convolution along $\oplus$ (Levi)	multiplication weighted by $q^{k(n-k)}$
Hall product (parabolic, §9)	Gaussian-binomial structure constants
derivative ∂ (hyperplanes)	Jackson D_q
shift σ (Def. 10.2)	$f(x) \mapsto f(qx)$
Ind_*, Res_* (Cor. 9.6)	multiplication, comultiplication
$q \to 1$	d/dx, binomials, ordinary generating functions

When the positions carry a group action the middle rung is not one-variable polynomials but the ring Λ of symmetric functions, and decategorification is the character map: the Frobenius characteristic for $\bigoplus_n \mathrm{Rep}(S_n)$, and Zelevinsky's for $\bigoplus_n \mathrm{Rep}(\mathrm{GL}_n(\mathbb{F}_q))$, in which Green's polynomials give the transition to the Hall–Littlewood basis (§8).

10.2 The programme

Programme. Every identity in the q- and (q,t)-theory of symmetric functions should be the shadow of a natural isomorphism of containers, and every container construction should decategorify to an identity. This is Joyal's species programme run at q: replace "bijection of finite sets" by "isomorphism of q-containers" and "generating function" by "q-series". The payoff sought is not new identities but *reasons* — a coefficient that looks arbitrary at the polynomial level should be forced at the container level.

10.3 A worked instance: the q-Leibniz rule

Definition 10.1 (Product of containers)**.** $(S \triangleleft P) \times (T \triangleleft Q) = (S \times T) \triangleleft \big((s,t) \mapsto Ps \oplus Qt\big)$, so that $[\![C \times D]\!] \cong [\![C]\!] \times [\![D]\!]$ pointwise and $f_{C\times D} = f_C f_D$.

Definition 10.2 (Shift)**.** σC is the container with $[\![\sigma C]\!]V = [\![C]\!](V \oplus \mathbb{F}_q)$. Explicitly $\sigma(S \triangleleft P) = \big(\sum_s (Ps)^*\big) \triangleleft \big((s,\varphi) \mapsto Ps\big)$: each shape s is repeated q^{d_s} times, positions unchanged. Hence $f_{\sigma C}(x) = f_C(qx)$, and σ preserves containers.

The Jackson derivative does not satisfy the ordinary Leibniz rule; it satisfies

$$D_q(fg) \;=\; D_q(f)\, g \;+\; \sigma(f)\, D_q(g), \qquad \sigma(f)(x) = f(qx),$$

an asymmetry that is unmotivated at the level of polynomials. It is forced at the level of containers.

Theorem 10.3 (Categorified q-Leibniz)**.** *For q-containers C, D,*

$$\partial(C \times D) \;\cong\; \partial C \times D \;+\; \sigma C \times \partial D.$$

Proof. Both ∂ and $\times$ distribute over $+$, so it suffices to take C and D with one shape each, of position dimensions d and e. Then $C \times D$ has the single shape of dimension $d+e$, so the left side has $\begin{bmatrix} d+e \\ d+e-1 \end{bmatrix}_q = [d+e]_q$ shapes, all of position dimension $d+e-1$. On the right, $\partial C \times D$ contributes $[d]_q$ shapes and $\sigma C \times \partial D$ contributes $q^d[e]_q$, again all of dimension $d+e-1$. The two counts agree by

$$[d+e]_q \;=\; [d]_q + q^d\,[e]_q. \qquad \square$$

Verified by enumeration in all 50 cases with $q \in \{2,3\}$ and position dimensions up to 4; in the same 50 cases the untwisted rule $\partial(C \times D) = \partial C \times D + C \times \partial D$ *fails*, so the shift is doing real work.

Remark 10.4 (Why the twist). Differentiating the second factor puts the hole in D; the C half of the structure has then already absorbed the extra dimension, and σC records exactly that. The asymmetry of the q-Leibniz rule is a statement about *where the hole is*, which is invisible once one has passed to polynomials. This is the programme working as intended.

10.4 Targets

identity	expected container statement	status
q-Leibniz	Theorem 10.3	**confirmed** proved, verified
q-Pascal	split the Grassmannian container on a vector	known geometrically
q-binomial theorem	a decomposition of σ^n	open
chain rule $\partial(C \circ D)$	no clean q-chain rule exists	open
Pieri, Littlewood–Richardson	Hall polynomials, §8	open
Macdonald $P_\lambda(x; q, t)$	needs a second parameter	§10.5

The chain rule deserves emphasis: containers *do* satisfy $\partial(C \circ D) = (\partial C \circ D) \times \partial D$, and there is no clean q-analogue of the chain rule for D_q. Either container composition deforms so as to repair it, or the failure has a structural cause. Both answers say something the polynomial picture cannot.

10.5 The two-parameter ceiling

Warning 10.5. Macdonald's parameters are (q, t) and *neither is the field size.* In the dictionary of §8, $t_{\text{Mac}} = q_{\text{field}}^{-1}$ and $q_{\text{Mac}} = 0$, since Hall–Littlewood is the $q_{\text{Mac}} = 0$ specialisation. We have one parameter, and it is the Hecke one.

The ladder of §8 suggests the rule: *one parameter per deformation direction in the base category.* $\mathbf{Vect}_{\mathbb{F}_q}$ has none and gives divided powers; finite $\mathcal{O}$-modules have one nilpotent direction and give Hall–Littlewood. Two parameters require a second direction, and there are three known ways to supply one.

Two nilpotent directions.

Positions in finite-length $\mathbb{F}_q[x, y]$-modules — the most container-native reading, "positions in a two-dimensional array with decay in each direction". This is Haiman's setting, via the punctual Hilbert scheme. The obstruction is sharp: $\mathbb{F}_q[x, y]$-mod has global dimension 2, so it is not hereditary; the Ringel–Hall product is still associative but Green's theorem fails, and polynomiality of the structure constants has no reason to survive. This is exactly why Haiman needed geometry rather than Hall-algebra combinatorics.

An elliptic curve.
Positions in $\mathrm{Coh}(E)$ for $E/\mathbb{F}_q$ elliptic. The Hall algebra is Burban–Schiffmann's elliptic Hall algebra, identified by Schiffmann–Vasserot with the spherical double affine Hecke algebra, whose eigenbasis on symmetric functions is Macdonald's. The two parameters are the two Weil numbers $\alpha, \bar{\alpha}$ of the curve, with $\alpha\bar{\alpha} = q$. This route works; container theory contributes only packaging.

Decoupling the two ends.
$[\![S \triangleleft P]\!]V = \sum_s \mathrm{Hom}(Ps, V)$ has *two* argument places and we have deformed one. Every parameter found so far comes from P; the value slot V has been left undeformed throughout. In the Macdonald operator $D_n = \sum_i \prod_{j\neq i} \frac{tx_i - x_j}{x_i - x_j} T_{q,x_i}$ the parameter t sits in the coefficients (the finite Hecke parameter, from the positions) and q sits in the shift operator T_q — which is Definition 10.2. That is the same split as in the double affine Hecke algebra, where t is the finite Weyl parameter and q the lattice-translation parameter. Currently $x = q^{\dim V}$ ties the shift to the field size; decoupling them is the open design question.

Honest assessment: the third route is the only one in which container theory does work rather than wrapping, and it is the least likely to be straightforward. Confidence that decoupling the value slot yields two parameters at all: perhaps 35%; that they are Macdonald's rather than an unrelated pair: lower.

11 Status of every claim

#	claim	prior	outcome
A	representation theorem (Thm. 4.1)	85%	**confirmed** — proved, plus 12 enumerations
B	fails when shapes linearise (Thm. 5.2)	85%	**confirmed** — proved; stronger than stated
D	derivative is Jackson D_q (Thm. 6.4)	—	**confirmed** — proved and verified
–	counting series (Prop. 6.1)	—	**confirmed** — verified by enumeration
C	two derivatives over $\mathbb{F}_q$	60%	**refuted** — $\mathbf{Vect}_{\mathbb{F}_q}$ is self-dual
C′	two derivatives over non-self-dual base	new	open; hand-checked on $\vec{A}_2$
H	Hall product over $\mathbf{Vect}_{\mathbb{F}_q}$ (Prop. 8.6)	—	classical
HL	*Day convolution = Hall product*	—	**refuted** — §8.2: counts splittings
HL′	two composition laws, one Hall–Littlewood	new	Thm. 8.12 classical; unverified
S	filtration depth $\perp$ self-duality failure	new	Table 1; middle row open
L	adjunctions lift to containers (Thm. 9.2)	new	proved; variance flips
I	$\mathrm{Ind}_* \dashv \mathrm{Res}_* \dashv \mathrm{Ind}_*$ (Cor. 9.6)	new	proved from Thm. 9.2
P	Day = Levi, Hall = parabolic (§9)	new	index computation; classical inputs
K	gap between $\mathbf{Cont}_q$ and Kuhn's $\mathcal{F}(q)$	—	open
M	Mackey as a distributive law (Q. 9.12)	new	open
LZ	categorified q-Leibniz (Thm. 10.3)	new	**confirmed** proved; 50 cases verified
G	lifting Macdonald theory (§10)	new	programme; one parameter so far

12 Related work

Abbott, Altenkirch, Ghani.
Categories of Containers (FoSSaCS 2003); *Containers: Constructing Strictly Positive Types* (TCS 2005). The theory being deformed.

Gambino, Kock.
Polynomial functors and polynomial monads. Source of the local cartesian closure requirement, hence of §2.

Kock.
Data types with symmetries and polynomial functors over groupoids. The species/quotient-container bridge; the structural model to imitate.

Kuhn.
Generic representation theory of $\mathcal{F}(q)$, the category of functors $\mathbf{finVect}_{\mathbb{F}_q} \to \mathbf{Vect}_{\mathbb{F}_q}$. By Theorem 5.2 this is the **Vect**-valued setting, in which container structure collapses; determining what $\mathbf{Cont}_q$ looks like inside the **Set**-valued analogue is open.

Zelevinsky.
$\bigoplus_n \mathrm{Rep}\big(\mathrm{GL}_n(\mathbb{F}_q)\big)$ as a positive self-adjoint Hopf algebra, degenerating to $\bigoplus_n \mathrm{Rep}(S_n)$ — symmetric functions, hence species — at $q = 1$. The representation-theoretic target.

Friedlander, Suslin.
Strict polynomial functors: the *other* linearisation, with positions a set and values a vector space.

Ringel, Hall.
The algebra of Definition 8.3; noncommutative exactly when the base has non-split extensions and no self-duality.

Kapranov, Vasserot; Schiffmann.
The Hall product as convolution on the stack of objects, along the span of short exact sequences — the construction that §8.2 says is needed in place of Day convolution.

Macdonald.
Symmetric Functions and Hall Polynomials, II–III: Hall polynomials, the isomorphism $H(\mathcal{A}_{\mathcal{O}}) \cong \Lambda$, and Hall–Littlewood $P_\lambda(x;t)$. The source for Theorem 8.12.

Green.
The unipotent character values $Q^\lambda_\mu(q)$ of $\mathrm{GL}_n(\mathbb{F}_q)$ as the transition matrix from power sums to Hall–Littlewood, degenerating to the character table of S_n at $q = 1$.

Langer.
PhD thesis (`~/git/lean/thesis-tex`): Macdonald-level generalisation of the Kawanaka identity, with the Hall–Littlewood case $q = 0$ as precedent. The nearest local expertise on the middle row of Table 1.

13 Next

1. **Question 8.16:** is the class of q-containers closed under Hall convolution? This decides whether §8 is a statement about containers or only about q-species.
2. **$\mathcal{O}$-position containers.** Enumerate both composition laws for $\mathbb{F}_2[t]/(t^k)$-modules, confirm Example 8.9, and check the Hall polynomials against Theorem 8.12.
3. **Settle Conjecture 7.4** by implementing $\mathrm{Rep}_{\mathbb{F}_q}(\vec{A}_n)$, and confirm the middle row of Table 1, isolating failure of self-duality as the sole obstruction to $\partial_{\mathrm{sub}} \cong \partial_{\mathrm{quot}}$.
4. Check that Theorem 4.1 is stable under change of base.
5. Determine whether $\mathbf{Cont}_q$ sits inside a **Set**-valued analogue of Kuhn's $\mathcal{F}(q)$, and measure the gap.
6. Ask whether a nondegenerate form on the position objects makes a q-container self-dual, and whether FFD Theorem 14.4 then classifies such structures into three families.

A Reproducing the computations

```
q-bool/code/qcontainer.py   definitions, enumeration, both checkers
q-bool/code/experiments.py  the six experiments reported above
q-bool/code/FINDINGS.md     raw results

$ cd q-bool/code && python3 experiments.py
```

All arithmetic is exact over prime fields; there is no sampling and no floating point. The exhaustive checker enumerates every function family and filters by naturality, making no use

of the Yoneda lemma; the generic checker enumerates candidates by their value on (s, id_{Ps}), verifies naturality of each in full, and separately confirms that every element of the source is $[\![C]\!](\varphi)$ applied to a generic.